"Control Is a Trajectory, Not a Point": Conceptualizing Control in Human-AI Co-Creativity

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Figure 1: Trajectory of a chef co-creating a recipe with an AI. Control is depicted as a dynamic, negotiable, and evolving relationship, where the arrows suggest that control is fluid and moves in either direction over time.

Abstract

Control is a critical yet underexplored concept in human-AI co-creativity and more broadly human-AI collaboration, where AI systems are expected to act as collaborative partners with creative autonomy. Existing frameworks for characterizing control remain limited and often fall short in capturing the tensions and complexities of co-creation dynamics. In this paper, we examine how experts conceptualize control and expect human-AI control dynamics by leveraging a recent framework on characterizing control as our theoretical probe. We conduct a semi-structured focus group study with nine experts in HCI, co-creativity, and AI. Our findings reveal that control is widely viewed as a dynamic, context-dependent construct that should adapt across different phases of co-creation, domains, and levels of trust in AI. Drawing on our findings, we propose a conceptualization of control along with actionable design implications for designing such AI systems. This work contributes to the literature on Human-AI collaboration, Computational Creativity, and HCI, advancing our understanding of control in co-creative human-AI partnerships.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Interaction techniques**; • **Applied computing** → **Media arts**; • **Computing methodologies** → **Artificial intelligence**.

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Keywords

Human-AI Interaction, Mixed-Initiative Control, Autonomy, Agency, Initiative, Balance of Control

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1 Introduction

Human-AI co-creativity is a distinct process where humans and AI contribute to generating creative artifacts or ideas as partners [38], potentially producing outcomes that surpass what either could achieve independently [46]. Co-creativity has gained immense momentum over the past five years, driven by the rapid rise of Generative AI (GenAI) systems. Commercial GenAI-powered co-creative systems such as ChatGPT (writing) [66], Suno (text-to-music) [92], and Media Foundation Models (text-to-video and text-to-image) [58] have pushed the boundaries of creative AI from the artist's toolbox [83] to everyday creators across society [4].

As a result, this evolution has driven researchers to develop frameworks that support equal partnership in human-AI co-creativity [18, 24, 54]. Unlike traditional AI that builds independent autonomous systems [89], co-creativity research seeks to design AI systems that function as co-equal creative partners, necessitating a balanced dynamics between humans and AI [35]. This shift in design philosophy builds upon classical creativity theories [77] and is reflected in Kantosalo et al.'s [37] five foundational dimensions for effective co-creativity: context (situational constraints), collective (shared effort), community (broader stakeholders), contribution (individual

inputs), and collaboration (the interactive process itself). Together, these dimensions frame co-creativity as a relational and situated process, moving beyond isolated acts of generation. Research highlights the evolution from "AI as a tool" to "AI as a partner" in co-creativity alongside the creation of new concepts such as "extended aesthetics," where technology enhances both human creativity and perception [53, Chapter 8]. However, as AI systems become increasingly capable creative partners, questions of control between humans and AI in co-creation become critical [48]. Central to this challenge are fundamental questions that have received limited attention: How to conceptualize control between humans and AI in co-creation? What are the dimensions and factors of control? How should control be distributed between human and AI partners?

Broadly defined, control is the act of restraining influence, directing influence, or ruling over another entity [57]. In a recent systematic literature review of human-AI co-creativity, control has been identified as one of the dimensions of relevance to the design of such systems [87]. In co-creativity, achieving the balance of control between humans and AI requires both agents to maintain creative autonomy while effectively sharing control [64]. This ensures users retain their sense of control in co-creation [26], while providing AI with creative autonomy [12] and maintaining a stable equilibrium in co-creativity [100]. However, the balance of control in co-creativity is shifting towards AI extremes, raising concerns about maintaining user agency and ensuring co-equal creative experiences [18, 55]. Researchers highlight that as co-creative AI gains more capacity and a range of actions, finding a balance between human control and agency while leveraging AI as a co-creator is essential [55, 102]. While a few existing theoretical frameworks attempt to characterize control in human-AI co-creation, their empirical validation remains limited, resulting in a persistent gap between theory and practice. Furthermore, little is known about how practitioners themselves conceptualize control or their expectations regarding control distribution between humans and AI during co-creation. In light of this, our research is guided by the following research question:

> *How do experts conceptualize control, and what are their preferences regarding control distribution between humans and AI in co-creative processes?*

To address our research question, we conducted a semi-structured focus group with nine experts in co-creativity, HCI, and AI. As a probing tool, we utilized MOSAAIC, a recently proposed framework designed to characterize and balance control in human-AI co-creation [32]. In line with Dalsgaard's concept of bridging concepts in HCI [8], our use of MOSAAIC served as a theoretical probe to foster meaningful dialogue between theory and practice, enabling us to critically examine how abstract constructs of control resonate with experts' lived experiences and expectations. We selected MOSAAIC after reviewing existing frameworks on control within the literature, identifying it as the most recent and comprehensive model for examining control dynamics in co-creation. Building on our findings, we offer a more grounded conceptualization of control in co-creativity, capturing expert perspectives on preferred distributions of control dimensions. Our work makes three contributions:

- **Empirical insights from domain experts on control preferences:** We contribute rare empirical data from domain experts on how they conceptualize and prefer to distribute control in human-AI co-creativity, addressing a critical gap between theory and practice.
- **Conceptualization of Control:** We offer a three-level conceptualization of control (micro, meso, and macro) that is grounded in empirical data, advancing theory in co-creative system design.
- **Actionable design guidelines:** We provide concrete implications for designing control in co-creative systems and outline future research directions in co-creativity and human-AI collaboration in general.

Overall, this work contributes to the growing body of literature in HCI and co-creativity by refining theoretical models through practitioner insights and laying the groundwork for future research into human-centered and actionable co-creative AI design.

2 Related Work

2.1 Human-AI Co-Creative Paradigms

The evolution of co-creative AI can be traced from its beginning in traditional creativity support tools (CSTs) [34, 68, 70, 101], that served as passive assistants to enhance human creativity without contributing directly to content generation, to more active AI partners that actively co-create with humans [12, 31, 44, 47]. In co-creative systems, both humans and AI actively participate as creative partners in the generative process, forming a collaborative dynamic that differs from CSTs [10, 11, 35]. Accordingly, different co-creative interaction paradigms have emerged to characterize partnerships. Mixed initiative co-creation (MICC) is often used as a substitute term for co-creation in the literature [74, 98], although MICC specifically emphasizes the alternating control and initiative between humans and AI [16, 45, 98]. Studies in human-AI communication show the importance of both the human and AI being able to initiate the process of co-creation and alternate between these two states [71, 72].

Unlike general human-computer interaction (HCI), human-AI co-creation creates a more complex relationship between humans and AI as 1) AI actively contributes and collaborates in the creative process, 2) AI takes on the human-like roles of partner, evaluator, and generator [40, 73], and 3) AI creates novel content which is blended with the user's contribution [76]. Designing co-creative systems has unique challenges due to the spontaneity of the interaction between humans and AI [13, 39]. Such complex dynamics between humans and AI require both parties to have some control and creative autonomy over the co-creation, while also allowing humans to feel ownership and agency [33]. Recent interaction frameworks in co-creativity also emphasize user control in co-creation. Moruzzi et al. [63] show that users' perception of control significantly shapes their co-creative experience. Similarly, Davis et al. [14] highlight balancing autonomy and control as a core design principle in their co-creative contexts. As such, we delve into the concept of control in co-creativity in the next subsection.

2.2 Control in Human-AI Co-Creativity

Control is a crucial yet under-explored aspect of human-AI co-creativity. A recent systematic review of human-AI co-creativity identified control as one of six core dimensions of co-creativity [87].

The authors stated that control is important to reveal how much involvement users want from the AI and how much of the creative process they prefer to retain. Moreover, control in co-creative systems differs fundamentally from control in traditional human-computer interaction. While conventional HCI emphasizes human control over computational tools, co-creativity requires a more nuanced understanding of shared control where both human and AI agents exercise creative autonomy [32]. However, in the co-creation literature, control is a loosely defined concept, with diverse definitions and overlapping terminology.

In the literature of co-creativity, control has been defined differently across different research contexts. Issak et al. [32] define control as the power to determine, initiate, and direct the process of co-creation. Muller et al. [65] define control as the tactical means of achieving a goal, and Singh et al. [87], borrowing the definition of control from UX design, define control as the degree to which users can direct, influence, or override the system's actions to achieve their goals. However, as control is nascent in the field of co-creativity, a multitude of terms are used to describe what control entails in the process of co-creation. We discuss the two most pertinent terms, autonomy and agency, below.

Autonomy is defined as the ability to choose creative actions [12] and has been identified as one of the five pillars of co-creative AI [12]. Agency is described as the capacity to act meaningfully [56], and in computational creativity research, it is defined as the "amount of autonomy" an entity possesses [42], hence, a higher level concept of autonomy. These three concepts—control, agency, and autonomy—are often conflated, where the loose or variable definitions of control in co-creation literature largely stem from their complex interrelationship. Different scholars emphasize diverse facets, leading to fragmented and sometimes contradictory conceptualizations [93]. For instance, a taxonomy of robot autonomy in Human-Robot Interaction (HRI) mentions "shared control" as an associated and subset terminology for "shared autonomy" [41]. Kadenhe et al. [33] argue that human-AI co-creation requires structured frameworks to facilitate the balance of AI autonomy with human control. This fragmentation underscores the need for characterizing and conceptualizing control in human-AI co-creation.

Understanding and properly designing control is not merely a technical consideration; it is an ethical and experiential one. When AI dominates or overrides human input, it risks reducing humans' creative engagement and eroding originality and skill [86]. Methnani et al. [59] argue that meaningful human control is necessary to ensure accountability, responsibility, and transparency in AI systems. Properly designed control structures also contribute in ways that leverage human strengths and respect creative intent [30]. Control is not only a practical design requirement but also a foundational principle for responsible and successful co-creative partnerships in human-AI systems [76].

Unsurprisingly, this significance is reinforced by empirical studies. Lawton et al. [44] demonstrated this complexity through Reframer, creating interface variants that prioritize either "emergence and system agency" or "control and user agency," revealing that control needs to be configurable. Research posits control as the transition point between human and AI contribution [50], where, in a novice-AI music co-creation, the AI is designed to take on more control during routine tasks but cede control to the human during critical or complex situations [82]. Moruzzi & Margarido [62] argue that control modulation between users and AI is a key dimension for optimal co-creative processes, emphasizing actionable user-centered design strategies to enable this flexibility. Singh et al. [88] found that "systems offering high user control lead to greater satisfaction, trust, and a stronger sense of ownership" [88].

Despite these insights, the field still lacks a cohesive, operationalized understanding of what control means in co-creative systems, especially from the perspective of domain experts. As Hillebrand et al. [27] argue, conceptual inconsistencies across disciplines, ranging from HCI to AI ethics, highlight the urgent need for integrative frameworks that account for interactional and systemic dynamics. Zhang et al. [104] also demonstrate through a comprehensive review of 134 HCI papers that understanding control mechanisms is paramount for effective agency distribution in co-creative systems. To address this gap, our study focuses on understanding how experts conceptualize and prefer control in human-AI co-creation.

3 Methodology

To address our research question, we began by conducting a literature search to identify an appropriate theoretical framework for characterizing control in human-AI co-creation. This framework served as a probing tool for our subsequent empirical investigation of control. Probes have the ability to act as "embodied questions" that exist in a co-creative, empathic, and shared context [96]. In design theory, there is a growing need for bridging the theory-practice gap to make conceptual ideas tangible and discussable [8, 23, 69]. In our work, we employed a theoretical framework as a probe to bridge this gap, prompting participants to articulate, examine, and challenge their understanding of control. This approach aligns with Boehner et al.'s [2] categorization of probes in HCI—specifically the "Probe as Participatory"—through the use of a theoretical framework for participants to interpret and explain their own practices.

Having identified our probe, we then conducted a focus group with experts in co-creativity, HCI, and AI to explore their conceptualizations and preferences regarding control distribution (RQ). We utilized the selected theoretical framework to scaffold the discussion. Finally, we performed qualitative analyses on the collected data to derive empirical insights. The detailed methodology is outlined in the following subsections, and a summary of the procedure is illustrated in Figure 2.

3.1 Literature Review for Selecting a Theoretical Framework as our Probing Tool

To identify structured models of control in human-AI co-creation that could serve as a lens for our study, we conducted a literature review. In this section, we detail the literature search process.

We began with a preliminary review of the literature, drawing on the research team's expertise on the subject and findings from recent publications. The purpose of this initial review was to develop an informed understanding of the existing control landscape and to identify the formal keywords needed to structure our systematic literature review (SLR). This preliminary review revealed that frameworks on Mixed-Initiative Interaction [99], Levels of Autonomy [61], and Human-Robot Interaction (HRI) [41] largely dominate the discussion surrounding control. Most relatedly, Hardin

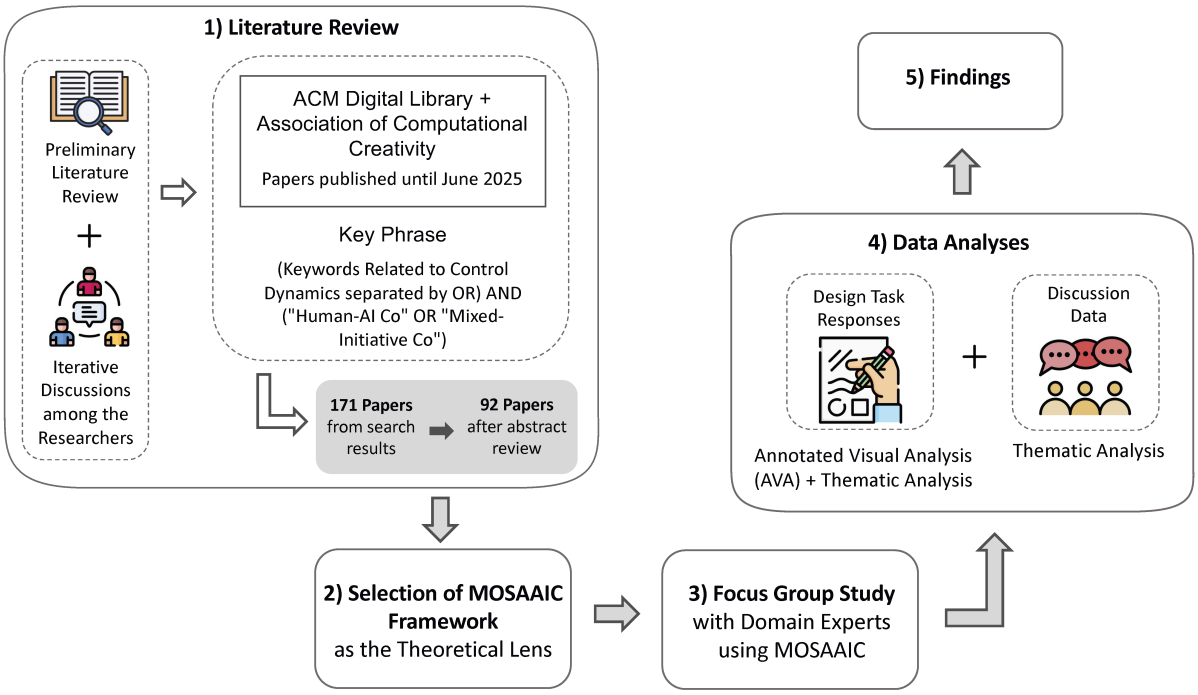


Figure 2: Research Methodology

and Goodrich [25] present Mixed-Initiative Control (MIC) in HRI as a perspective for managing workload in large-scale robotic teams, showing it outperforms both adaptive autonomy (agent-controlled) and adjustable autonomy (human-controlled). Although we identified several autonomy frameworks within HRI, they primarily address implementation-oriented or mechanics-oriented control, such as robot actuation and physical task execution, which do not adequately extend to creative contexts or to interactions with disembodied AI agents where control is a shared and complex process. Hence, our objective was to capture literature explicitly focused on control dynamics, control frameworks, or broader notions of control in human-AI collaboration.

To structure our search, we began by formulating a set of search terms through the preliminary literature review and iterative discussions among the research team, informed by prior work and recent publications across HCI, AI, and creativity. We developed a composite keyphrase that combined two main components: (1) a bucket of keywords related to control dynamics—referred to as the Control Bucket, and (2) keywords related to co-creativity. To ensure broad coverage, we included co-creativity terms such as "Human-AI co" to capture variations like co-creation, co-creativity, co-design, and co-dancing. Additionally, recognizing that "Mixed-Initiative Co-Creativity" is a widely used framing in the literature, we included the phrase "Mixed-Initiative Co" to capture those related works as well. The final query format was: **(Control Bucket keywords separated by OR) AND ("Human-AI Co" OR "Mixed-Initiative Co")**.

We executed the review across two primary databases: the ACM Digital Library (ACM DL[1]) and the Computational Creativity archive curated by the Association for Computational Creativity (ACC[2]). ACM DL was selected for its comprehensive coverage of HCI, AI, and co-creativity research. We considered ACC due to its strong relevance to our research focus. For ACM DL, we used its advanced search functionality to conduct structured queries. In contrast, because the ACC archive lacks such functionality, we developed a custom Python script to systematically query the site using our curated keywords. Our search, which included publications up to June 2025, yielded 171 unique results (132 papers from ACM DL and 39 from ACC). We limited the dataset to full-length peer-reviewed articles, excluding posters, tutorials, or extended abstracts to ensure scholarly rigor. For the initial screening, the first co-authors jointly reviewed abstracts of a few papers to calibrate inclusion criteria: papers had to directly or indirectly discuss control dynamics, frameworks, or related constructs in human-AI co-creative settings. Once agreement was established, the remaining abstracts were screened independently. This process resulted in a refined corpus of 92 papers (69 from ACM DL and 23 from ACC) deemed relevant to our goal (see supplementary material).

We looked for conceptualization of control in these 92 papers. While some prior work discusses related ideas such as agency, initiative, or authorial intent, these efforts are often fragmented and do not offer a formal model for understanding control, as we noted in Section 2.2. Existing research typically mentions control in passing or focuses on technical implementation rather than unpacking the

[1]https://dl.acm.org
[2]https://computationalcreativity.net/

concept in a way that is useful for guiding design in co-creative systems. For instance, Dang et al. [9] conduct a study to calibrate users' preferences for choice over control in narrative writing, Yang et al. [97] examine preferred level of control while transitioning between individual and collaborative learning activities, Lawton et al. [44] develop two versions of a co-creative AI called Reframer with varied levels of control and agency to explore how they influence user experience, Zhang et al. [103] develop VRCopilot to augment user controllability via mixed-initiative interaction, Liu et al. [49] discover that AI processing delays allowed users to enhance their sense of control, and Hoque et al. [28] develop interactive visualizations to help writers retain a sense of control and ownership of co-created text. In contrast to these studies, we found MOSAAIC (Managing Optimization toward Shared Autonomy, Authority, and Initiative in Co-creation) as the first framework to explicitly characterize control in human-AI co-creation, offering a structured and holistic perspective that breaks control down into three key dimensions and discusses how it may be dynamically distributed between collaborators over time [32]. From the SLR, we did not find any other conceptual frameworks for control, and instead, we found individual studies that touched on control or examined aspects of it without offering a formal model. Therefore, we chose this framework as our probing tool and describe its composition below.

3.2 MOSAAIC: Framework for Control Characterization in Human-AI Co-Creation

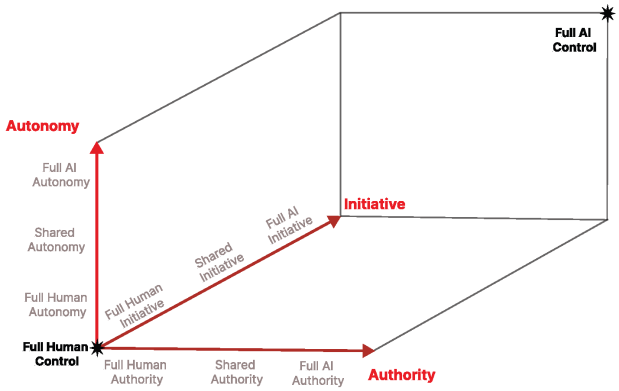


Figure 3: MOSAAIC Framework (Image from paper [32])

MOSAAIC characterizes control along three dimensions: ***autonomy***, ***initiative***, and ***authority*** (see Figure 3).

(1) *Autonomy* refers to an agent's ability to choose its own creative actions. MOSAAIC defines three levels: full human autonomy (AI acts only on direct commands), full AI autonomy (AI generates outputs independently), and shared autonomy (humans and AI both exercise creative freedom).
(2) *Initiative* refers to the capacity to proactively drive the creative process. In MOSAAIC, this ranges from full human initiative (AI only responds to prompts), to full AI initiative (AI acts without prompting), to shared initiative, where both human and AI dynamically initiate and respond in a collaborative exchange.
(3) *Authority* refers to decision-making power in the creative process. In MOSAAIC, full human authority places the AI in a purely supportive role, while full AI authority allows the system to direct the process independently. Shared authority enables mutual negotiation, where both human and AI shape creative decisions collaboratively.

We introduced the MOSAAIC framework during our focus group as a probing tool to stimulate discussion. This approach allowed us to bridge theory with practitioner insight and refine the conceptualization of control based on empirical understanding.

3.3 Focus Group

To identify how domain experts conceptualize control and their preferences regarding control, we conducted a focus group study. In this section, we detail the study methodology.

3.3.1 Participants. We used snowball sampling to recruit experts attending an international conference on co-creativity. This resulted in nine experts with a wide range of expertise and diverse backgrounds, including HCI, Computational Creativity (CC), AI, and the Arts. We verified expertise during introductions, where participants described their backgrounds and relationships with co-creativity research. Participants represented a variety of domain expertise as shown in Table 1. Our sample included both senior researchers who transitioned from related fields (P1, P7) and junior researchers who are building careers specifically in co-creativity (P6).

Table 1: Focus Group Participants

ID	Occupation	Domain Experience	Years
P1	Professor	Software engineering	20
P2	Professor	LLMs and audio production	10
P3	PhD student	ML and HCI	4
P4	PhD student	Music, HCI, IT, and arts	10
P5	PhD student	NLP and linguistics	6
P6	PhD student	CC and Co-creativity	5
P7	Professor	Writer, artist, philosopher	30
P8	Master's student	Applied mathematics	1
P9	Master's student	CC and Computer science	2

3.3.2 Study Procedure. One member of the research team facilitated the focus group in person with the participants, and two other members joined the focus group virtually as facilitators. The focus group study lasted approximately 2 hours, divided into four main phases. The procedure is detailed in Figure 4 below.

The *first phase* began with an introduction session where all participants introduced themselves and their areas of expertise and participated in an ice-breaker activity on co-creation. This session lasted approximately 10 minutes. The *second phase* was followed by a presentation by the researchers (see supplementary material). They described MOSAAIC, along with its three dimensions which took around 25 minutes. In the *third phase*, participants completed four design tasks, which lasted around 30 minutes in total. The first three design tasks comprised the three dimensions of control according to MOSAAIC: autonomy, initiative, and authority; the last

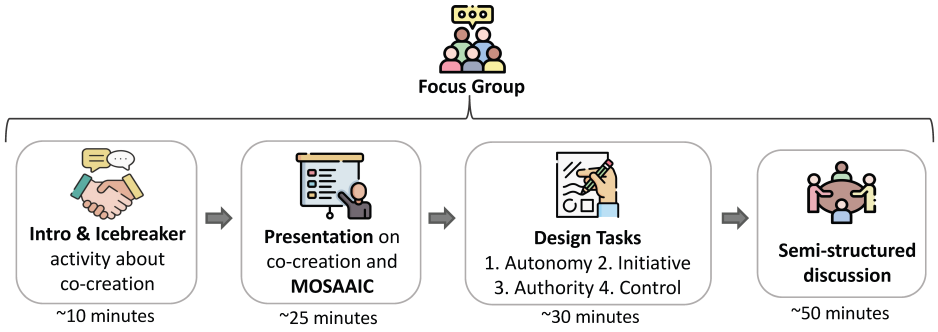


Figure 4: Study procedure of the focus group.

design task combined all three. Prior to each task, the researchers provided definitions and examples of each dimension. Each design task was facilitated by the following template:

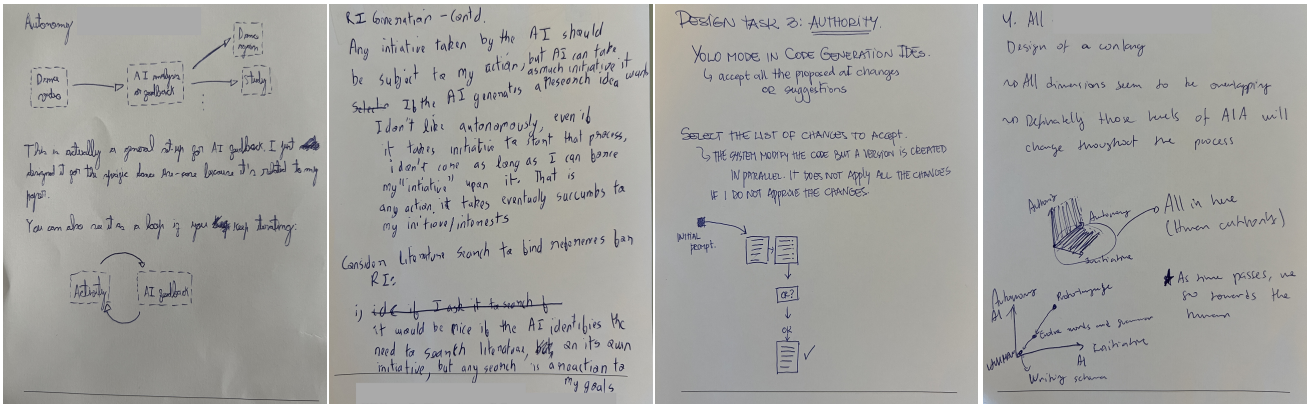
- *Brainstorm the design of a co-creative system where you see the balance of [a dimension or combination of all dimensions] between humans and AI.*
  - *You can pick any domain for the design task*
  - *You can sketch, scribble, annotate and/or write down your design ideas. Include descriptions of your design to complement visuals if possible.*
  - *You will have [6 or 8] minutes for the design task.* [6 minutes for the first three tasks and 8 minutes for the final task, based on estimated times identified in pilot studies.]

The *fourth and last phase*, included a semi-structured discussion, which took approximately 50 minutes, participants were invited to discuss their preferences and rationale for the distribution of autonomy, initiative, authority, and control in human-AI co-creation. We also asked about their ideal balance, trade-offs, and other factors or aspects of co-creation that they discern to influence how control should be shared between humans and AI. The study was recorded and transcribed using Zoom's transcription feature.

3.3.3 Data Collection & Data Analysis. We collected two main types of data from the focus group: (1) **Design Task Responses** and (2) **Discussion Transcript**. We collected a total of 35 handwritten design artifacts from participants, as P9 left after three design tasks due to time constraints. Each response was photographed and digitized (Figure 5).

We employed qualitative methods to analyze both the design task responses and the discussion transcript (see supplementary material). During the design tasks, we asked participants to "sketch, scribble, annotate and/or write down" their design ideas on their sheet of paper. As such, we received textual and visual responses. The design task responses were analyzed using a combination of Annotated Visual Analysis (AVA) [90] for the visual content and thematic analysis for the accompanying text. The discussion transcript underwent a separate thematic analysis. Zoom transcripts were manually edited by the research team to ensure accuracy and fidelity. Both thematic analyses followed Braun and Clarke's six-phase approach [3] and an inductive, open coding process informed by Saldana [81].

Design Task Analysis: For the visual data, we followed AVA's five-step process: *collect*, *collate*, *annotate*, *group*, and *confirm*, which allowed us to systematically examine visual artifacts such as sketches,



(a) Autonomy **(b) Initiative** **(c) Authority** **(d) All/Control**

Figure 5: Sample Design tasks submitted, each sample sourced from a different participant. From Left to Right - (a) Autonomy | (b) Initiative | (c) Authority | (d) All/Control

scribbles, and annotations. For the textual data, we applied Braun and Clarke's six-phase thematic analysis approach, guided by inductive, open coding practices [3, 81]. Two researchers first performed inductive coding on eight design tasks (22% of the data) to establish rigor and align coder variability for both analyses [78, 85]. Initial intercoder reliability was assessed using Cohen's Kappa ($\kappa = 0.75$), indicating substantial agreement. We then met to discuss discrepancies, refine codes, and reach consensus on the coded sample. Using the refined codebook, we independently coded the remaining 78% of the data in a split fashion and reconvened to establish the themes. We then reduced and categorized 59 codes using Miro, a collaborative digital whiteboard, resulting in 9 themes.

Discussion Transcript Analysis. We conducted a thematic analysis on the discussion transcript. Two researchers first performed inductive coding on 20% of the transcripts to establish rigor and align coder variability. Initial intercoder reliability was assessed using Cohen's Kappa ($\kappa = 0.70$), indicating similar substantial agreement. We then met to discuss discrepancies, refine codes, and reach consensus on the coded sample. Using the refined codebook, we independently coded the remainder of the transcript. This yielded 57 unique codes, which were later grouped into 10 themes through collaborative synthesis in Miro. After both analyses were complete, the authors collaboratively reviewed and integrated the findings. The 19 themes from the design tasks and discussion were distilled into *9 final themes*, organized under *3 overarching high-level categories*. These themes revealed insights regarding how practitioner perceive control, their needs, and emergent expectations of human-AI control dynamics. We present our themes in the next section.

4 Results

We present the findings from our qualitative analyses, organized into nine key themes. As shown in Figure 6, themes are grouped under three overarching categories that structure and contextualize our analysis. While the first category is influenced by the three dimensions of MOSAAIC, all three categories reveal novel findings about experts' conceptualization and preferences of control. The categories are:

- **Negotiating Control**: Captures how participants approached the distribution of control between humans and AI across the dimensions of MOSAAIC. This theme surfaces the tensions, trade-offs, preferences, and novel synthesis that experts expressed when co-creating with AI.
- **Building Control**: Focuses on mechanisms for realizing control in co-creativity. This includes the need for contextual adaptation of control, user customization to regain, share, or delegate control, and ethical guardrails.
- **Envisioning Control**: Captures participants' conceptualizations of balanced control dynamics, emphasizing a balanced partnership where AI can support, challenge, and grow alongside humans through productive friction and mutual empowerment.

Each category contains three themes. Table 2 provides a detailed summary of the themes, including the number of supporting codes, a definition for each theme, and example quotes.

4.1 Negotiating Control

This category captures how experts *negotiate* the distribution of control between human and AI agents during co-creation. Prompted by the three MOSAAIC dimensions—autonomy, initiative, and authority—participants expressed a range of expectations, preferences, and tensions that shaped their views on how control should be balanced.

4.1.1 ***T1: Categorizing Authority***. All participants wanted **full human authority** over co-creation. Participants were reluctant to cede their authority to AI, citing concerns that full AI authority could be dangerous (P2), hinder the creative process (P5), and lead to a loss of ownership (P3). P3 explicitly stated: "*[AI] is always*

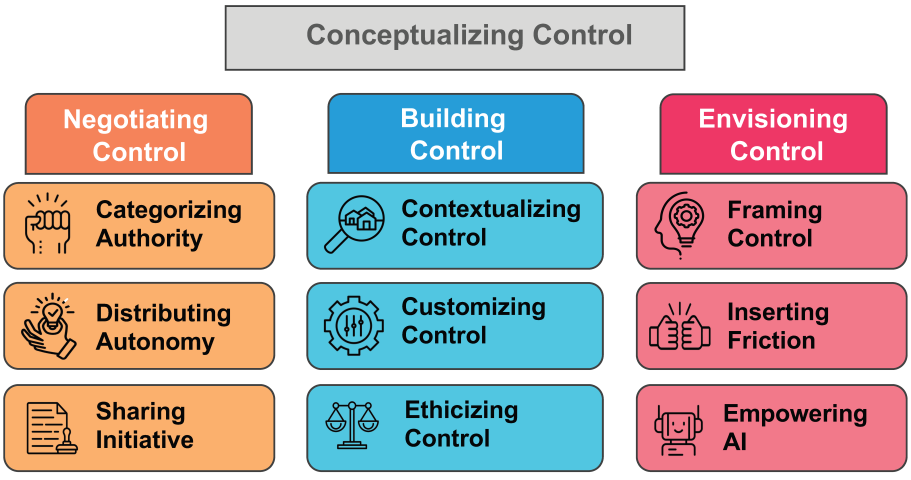


Figure 6: Results of qualitative analyses: overarching categories and corresponding themes related to conceptualization of control.

Table 2: Theme descriptions and example quotes.

Key	Category & Theme	Count	Description	Quote
	Negotiating Control			
T1	Categorizing Authority	20	How *decision-making ability* is distributed, with a strong preference for full human authority.	"*You're protecting the creative process with that authority.*"
T2	Distributing Autonomy	11	How *the ability to choose actions* is distributed, with preferences for granting AI autonomy with humans having control over when and how.	"*I would like the AI to do a lot of stuff autonomously.*"
T3	Sharing Initiative	12	How *the ability to proactively contribute* is distributed, with a general sense that AI initiative is welcome within boundaries.	"*AI should have full initiative to be able to design, provide, and implement* [...]"
	Building Control			
T4	Contextualizing Control	11	How control dynamics are realized depends on the creative phases, trust, domains, user type, etc.	"*Full AI authority at the beginning but at the end, full human authority*"
T5	Customizing Control	8	How humans should be able to adjust the control dynamics according to user goals and needs.	"*It would be nice to have a customization tool for flexible dimensions.*"
T6	Ethicizing Control	9	How to insert ethics into control dynamics, both to account for guardrails for AI but also to deal with human biases.	"*A guardrail [...], you can't make jokes about these subjects.*"
	Envisioning Control			
T7	Framing Control	18	How to frame control in the context of co-creation, with the general sense that it is a dynamic multifaceted concept.	"*Control is a trajectory, not a point*"
T8	Inserting Friction	13	How AI should provide a counterweight to human creative input, through creative friction.	"*Conflict drives creativity.*"
T9	Empowering AI	14	How the AI should be empowered to meaningfully contribute as a partner so it becomes truly a co-creation.	" [...] *AI can actually become your teacher.*"

subject to my authority, [where] in some sense, my use of it is subject to my goals." In this regard, participants presented their vision with two modifications to the presentation of authority in MOSAAIC: levels of authority and a spectrum of authority.

For **levels of authority**, participants devised hierarchical categorizations such as macro and micro (P3), and inside and outside (P6). These categorizations provided **higher-level authority to the human and lower-level authority to the AI**. Participants asserted that AI systems may be given authority to decide, control, create, and implement, but should never surpass human authority. As P3 explained: "*[AI] might be making minor decisions, but those minor decisions don't matter if my major decision overrides those.*" Attending to the co-creative process, P3 noted that human authority helps direct creative direction: "*[when] the AI is not going in a creative direction, I should stop it.*" P6 reinforced this, stating "*you're protecting the creative process with that authority*" by navigating the creative process to your liking.

For the **spectrum of authority**, participants presented limitations to authority distribution in the MOSAAIC framework, citing that there's a gray area that is not captured by the current three-stage spectrum of full AI, shared, and full human authority. Such as P1 suggested that this spectrum currently does not capture. They proposed a more **granular spectrum with five stages**, including *mostly AI* or *mostly human authority*, and one that positioned final authority to the human (Figure 7). Participants also added that the spectrum would be beneficial to the co-creative process.

However, most participants **questioned authority as the appropriate terminology** for the dimension. Since full AI authority was unacceptable to their conception of co-creativity, participants proposed alternatives. P6 stated: "*I don't know about authority... maybe* ***authorization*** *might be better?*" P4 suggested "*authority*

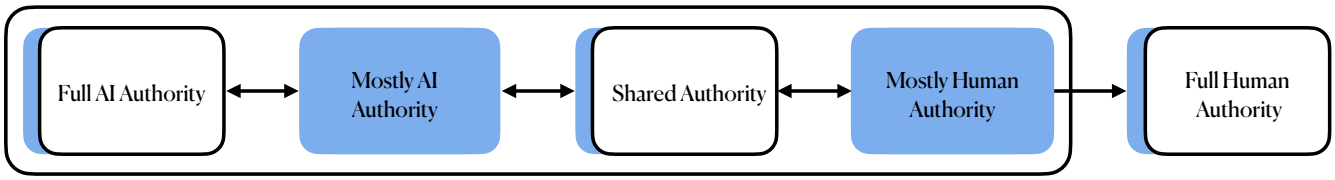


Figure 7: Participants proposed an authority spectrum with five stages. The three boxes in white are from the pre-existing MOSAAIC spectrum, and the two boxes in blue are the additions by the participants. The first four stages are interchangeable, whereas the spectrum always ends with full human authority.

as a different kind of ***evaluation***," where evaluation replaces authority by employing an assessment-based decision-making model. These alternatives avoid ceding the underlying attributes of authority (power and dominance) while advocating for continuous interaction and feedback toward a desired outcome. This finding challenges MOSAAIC as categorizing 'authority' as a dimension and advocates for participants' desire for authorization.

4.1.2 ***T2: Distributing Autonomy***. Participants widely supported granting AI creative autonomy, but consistently emphasized the need for user control over how and when that autonomy is exercised. Most **participants favored high AI autonomy**, particularly in augmenting human limitations. For example, P7 argued that "*the intelligent agent must be given full autonomy to surpass faulty social and technological systems terribly impaired by humanity's self-destructive drive.*" Similarly, P3, in their design artifact, illustrated AI autonomy as enabling the system to pursue creative directions independently, which is required in co-creation. Several participants preferred **high AI autonomy in early phases of co-creation**. For instance, P5 outlined a phased approach in their design response: full AI autonomy for tedious tasks in the beginning, shared autonomy in the generation stages, and full human autonomy in evaluation phases or at the end.

While AI autonomy was generally welcomed, participants stressed the importance of user-configurable AI autonomy. P3, in their design, supported high autonomy "*to do as much magic as possible*" but emphasized the ability to interrupt or override the AI. P4 similarly argued for the "*guarantee of user freedom any time*" and allowing the user to define the "*autonomy dynamics between humans and AI.*" P2 visualized interface controls for adjusting levels of autonomy as part of their design task response. P2 also proposed that **autonomy be assigned locally rather than globally**, suggesting that the AI's level of autonomy should vary across different tasks or stages of the creative process. While *global autonomy* applies the same level of AI autonomy system-wide, local autonomy allows for more fine-grained autonomy adjustments depending on the specific context or task. However, participants also raised concerns about **overly autonomous AI disrupting creative flow**. In this context, P8 warned that, in a co-creative writing setting, high AI autonomy might derail the collaboration: "*The AI would just write anything.*"

Despite broad support for AI autonomy, participants also delineated areas where full human autonomy is essential. P1, for example, asserted that rule creation and goal setting within the co-creative processes should remain under complete human autonomy. Several participants discussed **shared autonomy as a productive middle ground**. P2 stated that in a human-AI cake-baking scenario, "*With a shared autonomy, you might get a better cake.*"

4.1.3 ***T3: Sharing Initiative***. Participants emphasized the importance of initiative in human-AI co-creation. While most **participants preferred full AI initiative**, others stressed the need for user oversight, contextual sensitivity, and the ability to override AI initiatives. Participants supported full AI initiative in tasks that benefit from proactive AI contributions, such as generating ideas and offering suggestions. P7 remarked, "*AIs need to have full initiative to be able to actually design, provide, and implement interventions, installations, processes, and tools.*" P6 emphasized, "*AI should have initiative [ability] without the intervention of users.*" However, some participants raised concerns about the potential **downsides of unrestricted AI initiative**. For example, P2 explained that in a human-AI cake-making scenario, "*with full AI initiative, 50% of those cakes would be inedible.*" Others pointed to user discomfort, especially when AI suggestions were irrelevant or annoying. P6 described programming scenarios where AI suggestions break focus: "*It's very annoying... it's confusing for me.*" P2 agreed: AI initiative can be "*terrible.*"

Shared initiative was seen as a productive middle ground that fosters novelty and variation in outcomes. P2 noted that "*shared initiative produces variation in outcome,*" while P6 described a preferred 50/50 split in initiative for a dance choreography scenario. In the group discussion, P6 added that, in a human-AI cake-making co-creation, "*shared initiative or shared prompting might give you a better cake.*" A recurring theme was the need for users to be able to override or intervene when the AI takes initiative. P3 explained, "*I don't care as long as I can bounce my initiative upon it. That is, any action it takes eventually succumbs to my initiative/interests.*" Participants also noted that the **AI initiative should evolve** over time based on its performance and the quality of its contribution. P8 stated that AI "*could gain a degree of initiative as time progresses,*" implying that AI might be granted more freedom to act as it demonstrates competence or alignment with the user's goals. P6 argued that AI initiative should be regulated based on the AI's purpose and the consistency and quality of its output as evaluated by the human collaborator.

4.2 Building Control

This category focuses on practical mechanisms for operationalizing control in co-creative systems, including contextual adaptability, user customization, and ethical oversight. The themes highlight the importance of balancing AI freedom with human oversight in ways that respond to specific contexts and evolving factors throughout the creative process.

4.2.1 ***T4: Contextualizing Control***. Participants emphasized that temporality is a crucial factor shaping control in co-creative systems. Participants advocated for the idea that human-AI control must evolve across different **phases of co-creation**. P1 envisioned a shifting dynamics of control over time: "*I will start with full AI authority at the beginning...I want to end very close with full human authority.*" Similarly, P4 emphasized that control dimensions should shift across different phases of the creative process, allowing dynamic adjustments as the collaboration evolves over time. P2 echoed this by describing co-creation as a process with "*checkpoints in each different stage until you get your final completed point.*"

Another crucial factor discussed was **trust**. Participants consistently linked trust with willingness to grant control to the AI. P2 remarked, "*If you gave AI full authority, that's asking for trouble,*" and added that one would "*give AI a greater degree of autonomy if you trusted the outcomes.*" Similarly, P1 questioned, "*Would you ever want to give real authority of your life to AI?*" P3 and P7 both emphasized that trust in AI is shaped by users' prior experiences with intelligent systems. P7 noted, "*It would depend a lot on our experiences, like the kind of intelligence we are following,*" suggesting that users' willingness to grant control to AI is influenced by how familiar, reliable, or aligned previous AI interactions have felt.

The **context and purpose** of the system were also seen as decisive factors. P4 remarked, "*Depending on the system's purposes, the developer can set each dimension as needed.*" P8 emphasized that control dimensions like initiative should adapt by "*time, purpose/context, and consistency/quality of AI contribution.*" P5 noted that developers could set the initial control parameters based on the system's intended purpose. Participants emphasized that AI authority must be carefully overseen, especially in sensitive or high-stakes contexts, and should only be granted in narrowly defined situations, such as evaluating user actions. The **creative domain** emerged as another contextual determinant. As participants completed their design tasks, choosing different domains (e.g., education, creative writing, or music), they exhibited varying preferences in control configurations. P4 observed that AI initiative might vary by domain, implying that a one-size-fits-all approach would be inadequate.

4.2.2 ***T5: Customizing Control***. Participants emphasized the importance of giving users the ability to adjust control dynamics. Rather than relying on static control structures, they envisioned systems that **allow users to dynamically modulate** control dimensions in alignment with their creative goals.

Participants envisioned tools that allow for flexible customization, particularly through interface components such as **sliders or knobs**. P4 captured this vision clearly: "*It would be nice to have a customizable tool for the flexible dimensions.*" Participants' design included sliders for adjusting levels of control dimensions, suggesting: "*The user should always be able to set these sliders for the three dimensions.*". P1 echoed this need for sliders for each dimension for granularity, noting that users should be able to override or redirect the AI, especially when they disagree with its outputs. P8 expanded on this concept by saying, "*One could envisage a leveled knob system, whereby a user could set the level of autonomy, initiative, and authority for the AI.*"

Participants also suggested that **different phases of co-creation might require different configurations** of control, and users should have flexibility in configuring control. As P4 noted, "*Depending on the UI principles and goals, different ways can be implemented for allowing the user to set [the sliders] at any time in the various phases of the creative process.*" P3 emphasized the importance of user intervention, suggesting that "*the user should be able to intervene AI.*" P6 also highlighted that users should have the right to "*choose or reject the AI contributions if they want.*" Participants widely emphasized the importance of aligning system control mechanisms with users' goals and preferences; for instance, P4 advocated for embedding human-centered design principles into co-creative systems to ensure that control dynamics reflect user intent and values.

4.2.3 ***T6: Ethicizing Control***. Ethics underlie the entire discussion on co-creative systems. Participants revealed **concerns about attributing control to the AI**. Participants imagined scenarios where AI refuses to cooperate (P1) or where the artificial actors surpass human actors and capabilities (P2). Most participants addressed the **need for guardrails**. In a hypothetical scenario of a co-creative system that makes jokes, P2 envisioned "*a guardrail in terms of that's a spoiler, you can't make jokes about these subjects.*" P1 emphasized severity: "*especially with human authority [when] thinking about ethical issues [and] dealing with humor.*" However, contrasting the above two discussions, P3 brought up a scenario where high AI authority can reduce human bias, where the "*AI should take authority to force you out.*" Participants addressed human fallibility, noting that **humans are not rid of their bias to assert control** and that AI can help address some of these preconceived biases.

Participants also highlighted that **AI lacks accountability and sensibility, even if trust in the system were to be achieved**. P3 explained: "*There is some contention that I can hold it responsible and can't hold you responsible in the same way,*" with P8 stating that: "*there is no 'IT' to be made accountable.*" Connecting the embedded human bias, values, and flaws in datasets, participants stressed the importance of data curation and appropriate training algorithms to maintain ethical development of co-creative systems that can mitigate the human biases that may be present in co-creative systems.

4.3 Envisioning Control

This category captures how participants conceptualized control, where co-creative AI is expected not only to support but also to challenge or provoke users to introduce creative friction. It includes perspectives on AI as a partner that contributes meaningfully rather than simply following human direction, revealing reflections on control.

4.3.1 ***T7: Framing Control***. Participants envisioned control in co-creation as a **dynamic, multifaceted concept** that must be

flexible across contexts. P1 described control as a temporal "*trajectory*" rather than a point that might begin with high AI control and move toward greater human control by saying, "*as time passes, control should go toward humans.*" In contrast, P7 proposed the reverse: control should gradually shift from humans to AI to enhance capability. P3 imagined a non-uniform conceptual space of control, describing it as a blob indicating areas of conceptual density. Participants also acknowledged potential differences between human and AI perceptions of control. For instance, P1 noted that both agents might believe they hold authority, leading to mismatched assumptions during collaboration. This complexity underscores the importance of establishing shared mental models of control for effective human-AI co-creation.

Participants emphasized the importance of **balancing control between humans and AI across the dimensions**. Several participants described the need for balance in their design task responses, with P2 pointing to the importance of finding a sweet spot in prompting between human and AI contributions. P4 and P2 emphasized that extreme distributions of control, entirely human or entirely AI, would undermine true co-creation. In this context, P4 noted, "*We need some balance, not the extreme (full human or full AI), because there is no co-creation.*" P7 suggested that this might entail humans gradually "*relinquishing control to the AI.*" P8 added that any balance must consider trade-offs between safety and efficiency.

While conceptualizing control, there was substantial discussion on the three dimensions of control according to MOSAAIC about **the dimensions being interconnected**, especially initiative and autonomy. P3, P5, and P6 each expressed difficulty distinguishing them. P5 said, "*All dimensions seem to be overlapping.*" P3 reflected, "*Initiative and autonomy are very correlated...imagine a Ph.D. student who is not at all autonomous but has very high initiative. What does that even look like?*" P6 observed, "If you don't have initiative, how are you going to authorize something or not?" P6 raised concerns about the alignment between authority and agency, prompting a call to clarify and sharpen the model's terminology.

4.3.2 ***T8: Inserting Friction***. Participants described the nature of interaction in co-creativity as one that need not always be smooth and harmonious, but rather one that introduces creative friction. P2 used the metaphor of a kitchen with opinionated cooks that come to a great dish at the end to capture the potential of AI that challenges humans. Two forms of interaction have been identified as valuable: adversarial interaction and interaction with feedback loops.

Participants highlighted the untapped potential of **adversarial human-AI interaction** in co-creative systems, an approach where control is not rooted in dominance but in mutual challenge and dynamic exchange. In current systems, unsolicited AI input is often seen as disruptive. As P2 noted, "*it pushes lots of suggestions continually as if you want to listen to them and you don't want to hear them.*" Rather than simply reducing AI intervention, participants proposed reimagining intrusiveness as productive friction, where disagreement and resistance from the AI could enhance the creative process. P2 added that, "*the warring aspect actually would make a better outcome*" with P1 summarizing the process as "*conflict drives creativity.*" To operationalize this interaction, P2 devised three strategies: evaluation, provocation, and experimentation. P2 explained: "*When you do something and you get evaluated... I can evaluate the AI, the AI can evaluate my results, this is provoking and adversarial behavior.*" They also proposed practical implementations, such as gamification, by envisioning AI as a player in a game that challenges you.

Participants also emphasized the need for **feedback loops** between human and AI. They envisioned iterative co-creation cycles where the AI offers ideas, the human evaluates and filters them, and the AI then adjusts based on this feedback. P3 illustrated this as a process where humans guide AI by setting parameters, evaluating outputs, and refining subsequent iterations. They also called for the inclusion of different interaction modes such as conversations separate from artifacts (chatbots), system designs prioritizing variety and quality, and feedback loops encouraging dynamic human-AI collaboration. P8 coined the term "*mature co-creation*" to describe stable systems that learn from continuous human interaction and tailor responses accordingly, whereas P6 suggested allowing AI to "*pass the turn without doing*" as feedback that would make AI feel sub-optimal and more like a human partner.

4.3.3 ***T9: Empowering AI***. Participants discussed ways to empower AI as a more capable creative partner, emphasizing both its functional capabilities and the quality of its contributions. For **AI capabilities**, participants imagined an AI that **guides, teaches, and orients** the human towards novel approaches and creative directions. P2 envisioned an AI that introduces possibilities beyond the user's scope: "*[AI goes] outside that process and says this is something that could be created.*" P6 highlighted the role of AI in co-creation: "*During the process of creation, it's interesting to have something that I didn't previously think of. It socially provokes.*" P1 imagined an inquisitive AI asking: "*Oh, have you tried this?*" introducing creative twists. These examples position the AI not as a subordinate tool, but as a partner. P2 encapsulated this vision: "*Then the AI can actually become your teacher.*" In such dynamics, control may temporarily shift to the AI, as it directs the co-creative flow with its guiding influence.

For **AI contribution**, participants scrutinized the quality of an AI's contribution in the co-creative process. When asked to elaborate on their thoughts on AI contributions, P3 explained that it is "*how well it co-intersects with your interests and goals*", noting that weak or misaligned outputs often led them to revert to fully human-driven creation. Participants preferred delegating a variety of tasks to the AI to complete autonomously, but only if the system's contributions met a certain standard—evaluating their usefulness before moving forward. Participants emphasized that the **consistency in AI contribution improves co-creation and trust**. Ultimately, meaningful contributions were seen as a prerequisite for sustained human-AI collaboration.

5 Discussion

Despite being identified as one of the foundational aspects of human-AI co-creation [87, 104], the concept of control remains under-theorized and empirically underexplored [32]. Existing literature often treats control abstractly or assumes it implicitly, leaving a significant gap in how it is actually experienced, negotiated, and operationalized by practitioners in real-world co-creative contexts. In this work, we sought to answer the question, *how do experts*

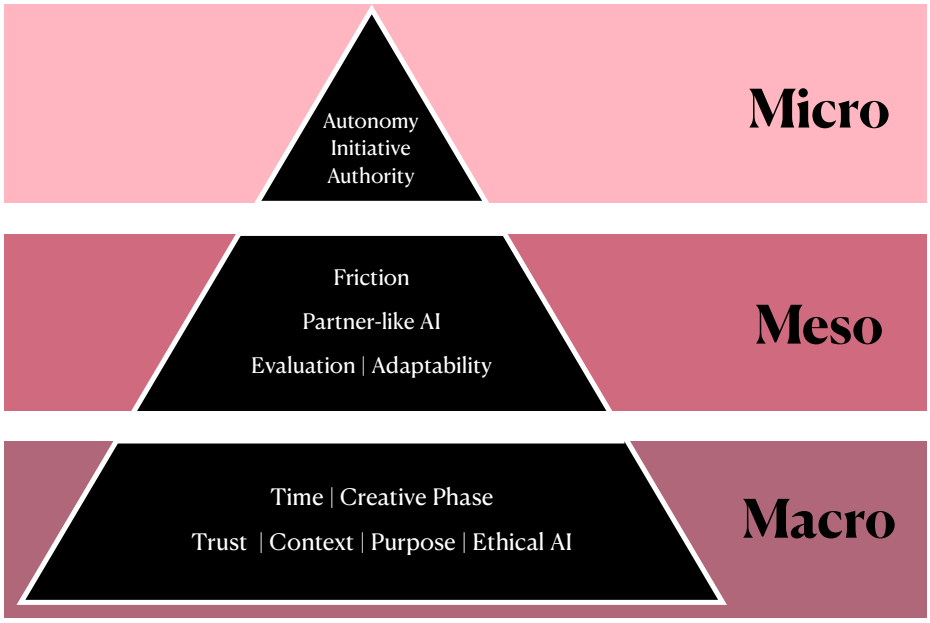


Figure 8: Conceptualization of control

conceptualize control, and what are their preferences regarding control distribution between humans and AI in co-creative processes. We outlined the findings revealing the nuanced expectations, tensions, and values that shape how control is conceptualized in practice. Below, we synthesize our key contributions. First, we present a **conceptualization of control**, based on our findings from the study. Then, we propose **actionable design strategies** for the development of human-centered co-creative systems. Finally, we outline our limitations and future work for this study.

5.1 Conceptualization of Control

Remy et al. [69] highlight the persistent theory-practice gap in HCI, emphasizing the need for frameworks that anchor abstract concepts such as control within real-world contexts. This critique resonates with existing control theories that often remain disembodied from lived experiences. Our study addresses this gap by offering a conceptualization of control in human–AI co-creation scaffolded by theory and grounded in expert reflections. The realized conceptual model of control is visualized in Figure 8. It presents three interrelated levels of control categorized into Micro, Meso, and Macro. The three layers in our visualization are fluid and dynamic, not isolated levels. Fluid negotiation among these levels must be supported, enabling shifts based on creative trajectory. We describe each layer in detail below:

5.1.1 Micro Level of Control. This level captures the moment-to-moment capabilities of human and AI agents to act or decide—such as autonomy, initiative, and authority. It reflects how operational control is distributed and exercised during specific phases of co-creation. Participants challenged the idea that autonomy, initiative, and authority are discrete and independently adjustable dimensions. Instead, they framed them as interdependent experiences, for example, asserting that it is not logical for a system to demonstrate initiative without some level of autonomy. This critique reveals that while the MOSAAIC framework treats these dimensions as orthogonal, they are often co-experienced and co-constructed. For instance, for *Autonomy*, participants wanted AI to handle low-level or tedious tasks independently, but preferred human direction for critical or evaluative moments. For *Initiative*, some participants favored proactive AI that suggests new directions, while others emphasized the importance of user primacy in ideation, advocating for AI to "hold back" until invited. For *Authority*, participants often reserved authority for humans, but also proposed contextual delegation, e.g., letting AI evaluate low-stakes outcomes or provide feedback loops that humans could accept or reject. These preferences, overall, highlight the interrelated nature of autonomy, initiative, and authority that was unexplored in MOSAAIC.

5.1.2 Meso Level of Control. This level is about the ongoing relational dynamics between human and AI throughout the creative exchange. It describes how control is realized in the flow of interaction, shaped by adaptability, evaluation, and friction. Most intriguingly, contrary to the assumption that optimal collaboration is always harmonious, participants emphasized the importance of creative *friction* in co-creation. Adversarial interaction, where the AI pushes back, disagrees, or challenges the user, was seen not as disruptive but as a vital mechanism to enhance creativity and engagement. Participants called for reimagining unsolicited AI suggestions not as intrusions, but as provocations that could drive deeper reflection and ideation.

Participants strongly advocated for a *partner-like AI* rather than a passive tool, echoing prior co-creative literature that emphasizes the value of partner-like AI in fostering shared creativity [36, 71, 75].

They also envisioned an AI that could teach, guide, and inspire, offering meaningful contributions to the process, rather than merely responding to commands. Aspects of this teacher-like AI have been a growing area of research in the HCI literature through the conceptualization of the "socratic method" in AI, i.e., a mode of interaction that simulates guided, inquiry-based learning and problem-solving through dialogue [1, 15, 19, 67, 84].

Adaptability of control emerged as a key expectation: participants sought systems that could flexibly adjust levels of control in response to evolving needs and goals. Additionally, participants proposed bidirectional *evaluation*, where both the human and AI assess each other's contributions, as a mechanism to improve mutual understanding, build trust, and guide when and how control should shift during collaboration. This level is an active area of study for HCI researchers.

5.1.3 Macro Level of Control. This level encompasses the broader temporal and ethical circumstances that shape how control should evolve. This includes factors such as trust, creative phases, purpose, and the context for co-creative system design. Participants viewed *trust* as an earned feature and emerged when AI made consistent, high-quality contributions, highlighting the importance of responsiveness and reliability in shaping long-term in human-AI relationships. Corbett and Le Dantec [7] posit trust as a dynamic value comprised of four elements: distance (uncertainty presented by distance), process (building trust through initiating, building, and retaining), sociotechnical components (artifacts, practices, and social arrangements), and design affordances (history, experience, expectation, and preservation). This conceptualization puts forth similar parallels for theorizing trust in co-creativity: distance (local or commercial system), process (initiating, choosing, and deciding), sociotechnical components (creative artifacts and creative practice), and design affordances (novice vs. experts).

Participants overwhelmingly framed control as contextual, shifting over time in response to creative phases, creative goals, purpose and context of the system. They emphasized that different *creative phases* of co-creation require different configurations of autonomy, initiative, and authority. Rather than assuming fixed levels of control for humans or AI, they envisioned control trajectories, for instance, with high AI autonomy to spark exploration, followed by increasing human autonomy as the user gains creative clarity. Others imagined the reverse: beginning with tight human oversight, then gradually granting AI more freedom as it proves capable. These temporal dynamics reflect how control evolves within and across creative phases, and posit the exploration of control dynamics within convergent and divergent phases of creativity as widely explored in tangential co-creativity literature [43]. In doing so, the *context* and the *purpose* of the task emerged as salient factors for determining responsive adaptations to control.

Ethical considerations were also central to this level. Participants expressed concerns about delegating too much control to AI systems—particularly in cases where the AI might act without moral guardrails, deliver inappropriate content, or lack accountability. Scenarios such as AI refusing to cooperate or overstepping creative boundaries (e.g., making offensive jokes) revealed a shared need for value-sensitive design that embeds ethical foresight and guardrails into the system. While some participants imagined AI reducing human bias by "forcing users out" of ingrained habits, others emphasized that AI's lack of sensibility and accountability poses fundamental risks.

5.2 Actionable Design Strategies for Co-creativity systems

In this section, we present design strategies that translate our conceptualization of control into tangible examples for building co-creative AI systems. While grounded in the context of creativity, these strategies extend to broader AI and HCI domains, offering guidance on how to balance control between human and AI agents across diverse applications. These strategies aim to guide researchers, designers, and practitioners in creating systems that support fluid, relational, and adaptive control dynamics across three levels. We outline four key design strategies informed by our findings in Table 3 below.

5.2.1 Operationalize Friction as Creative Provocation. Participants valued creative friction [T8] as a productive force in co-creativity. Recent work in HCI has already begun to push for this potential in human-AI collaboration. Gero et al. [21] propose "AI suggestion as challenge"; Villareale et al. [95] model "adversarial AI gameplay"; and Zhou et al. [105] present "AI as an opinionated colleague" utilizing their effects for adversarial tasks. Examples of implementing this design strategy include *provocative AI suggestions* that challenge user assumptions or introduce novelty and disrupt habitual creative patterns (e.g., "What if we reversed the tone of this piece?") [6]. Designers can also introduce *tunable friction levels* to provide users with options on how to calibrate how much challenge or provocation they would want (e.g., "inspirational," "critical," "provocative").

5.2.2 Embedding Feedback Loops for Evaluation and Trust. Evaluation emerged as a key mechanism for exercising shared authority, autonomy, and initiative [T1, T2, T3], implementing guardrails [T6], and building trust in co-creation [T9]. Rather than relying on fixed control settings, participants preferred to assess and selectively adopt AI suggestions—positioning evaluation as a dynamic enactment of control. In the dynamic conceptualization of control, feedback loops (meso level) embed evaluation that help users calibrate trust (macro level) and gradually negotiate creative responsibility with the AI (micro level). Examples of implementing this design strategy include *Accept/Modify/Reject buttons* for AI-generated contributions with contextual justification [20] (e.g., "The model suggests this based on your last 3 inputs"), *explanatory feedback loops* where the AI explains its rationale and allows users to critique its judgment [91] (e.g.,"I chose this style because it aligns with your prior image"), and *evaluation scaffolds*, such as structured critique prompts or reflection cards, to help users assess AI contributions with nuanced assessment [94] (e.g., "How does the color combination inform your design choice?").

5.2.3 Visualizing Control Trajectories. Participants conceptualized control as temporal and evolving, rather than static. In this regard, co-creative systems are primed to provide visual representations of how control dynamics shift over time across the three levels of control to foster reflection and responsiveness [T7]. Examples of implementing this design strategy include *control trajectory maps*

Table 3: Summary of actionable design strategies for co-creative systems

Design Strategy	Theme	Description	Examples
Operationalize Friction as Creative Provocation	T8	How friction emerges as a productive force for co-creativity by disrupting perceptions of *control*.	Provocative AI suggestions and Tunable friction levels [6]
Embedding Feedback Loops for Evaluation and Trust	T1, T2, T3, T6, T9	How *control* is earned and shared through the implementation of feedback loops that embed trust and evaluation.	Accept/Modify/Reject buttons [20], Explanatory feedback loops [91], Evaluation scaffolds [94]
Visual Control Trajectories	T7	How *control* dynamics are visualized as they shift over time and across the three levels of control.	Control trajectory maps [17, 80], Phase-sensitive prompts/UI cues [79]
Calibrating Dynamic Control Mechanisms	T4, T5	How *control* is adjusted before or during the co-creative process through dynamic configurable mechanisms.	Steerable interfaces [51], Role-switching features [60], AI Control Cards [29, 52]

that track changes in human-AI collaborative systems and *phase-sensitive prompts/UI cues* that signal when control handoffs are likely or beneficial, such as suggesting human authority during refinement stages. Both of these examples highlight beneficial additions to current visual displays in human-AI interaction dashboards [17, 80] that address and potentially adapt to the temporal aspects of human-AI collaborations [79].

5.2.4 Calibrating Dynamic Control Mechanisms. Rather than predefining control at the system or user level, co-creative systems should treat control as negotiable and configurable across creative phases, contexts, and user preferences [T4, T5]. Therefore, aspects of the micro level (control dynamics) and meso level (interaction dynamics) become configurable across the macro level (contextualization). Examples of implementing this design strategy include *steerable interfaces* [51] that enable the configuration of co-creative expectations prior to the creative process, similar to tuning musical instruments before a performance, where "getting lost in the flow" of the process is desired. Other configurable mechanisms include *role switching features* [60] (e.g., coach, collaborator, or evaluator) during the process of co-creation, as well as *AI Control Cards* (similar to design cards) [29, 52] that enable users to configure control dynamics based on the phase of co-creation (e.g., divergent vs. convergent).

5.3 Limitations and Future Work

We acknowledge the limitations in our study, which point to promising directions for future research. First, our recruiting strategy via snowball sampling from attendees at an in-person research conference may have introduced selection bias and limited representational diversity, a common challenge highlighted in human-AI studies [22]. Although we prioritized group diversity in expertise and background, we acknowledge that our sample size remained relatively uniform and small.

Second, and relatedly, our participant pool consisted primarily of expert designers and researchers. We purposely sought to engage with an academic community first to ground and build the foundation for conceptualizing control, and the insights of our participant pool, as documented here, were invaluable in articulating theoretical and design perspectives. However, future work should conduct empirical studies (addressing sample diversity and size) to understand whether the conceptualization and expectations of our experts extend to other experts and align with the needs and expectations of creative practitioners. Practitioners' experiences of control, in particular, would solidify the generalizability and transferability of the dynamic conceptualization of control.

Third, our use of the MOSAAIC framework as a theoretical probe shaped the direction and vocabulary of our investigation, as evident in the micro levels of control. While this choice was intentional—MOSAAIC is among the most recent and comprehensive frameworks for understanding control in AI co-creation—it seems to have influenced our findings. Using a different probe or multiple frameworks might have revealed different constructs or interpretations. Zwanenburg [106] indicates that using multiple frameworks or models provides stronger validity to construct or measure a concept. Future research should explore multiple, emerging conceptual models to develop a richer, more robust understanding of control in human-AI co-creation, for which we lay the groundwork with this work.

Finally, our findings suggest that control is not static but dynamic and fluid between the levels, where the proposed three-level framework of control offers a high-level and formative conceptualization. However, while our actionable design implications offer ways to translate our conceptualization into practice, our current framework does not yet offer granular operational definitions for transitions or interdependencies. As such, we have omitted transitions or arrows in Figure 8 until we have a granular understanding of the relationship between and in-between these levels. Research shows that the relationship between abstract conceptual models and practice is more complex than just instrumental logic, as models can induce dispositions that affect problem-solving [5]. Hence, future work should empirically investigate how control transitions occur across levels in real-time interactions and develop interaction techniques that facilitate such fluid negotiation.

6 Conclusion

In this paper, we set out to conceptualize control dynamics in co-creativity by examining how experts perceive and expect control to function. Using the latest framework as a theoretical probe, identified through a systematic literature review, we found that experts view control as dynamic, context-dependent, and evolving across creative phases. Rather than treating AI as a passive tool, participants favored partner-like AI that introduces creative friction and adapts to user goals. Our findings led to a three-level conceptualization of control, capturing distinct control constructs of capability, interaction, and context. Our findings underscore the importance of temporality, trust, and adaptability, pointing to the need for systems that support fluid negotiation of control. To help translate this conceptualization into practice, we proposed actionable design strategies to translate our findings and conceptualization of control. Our contribution bridges theoretical and practical gaps in existing literature and lays foundational groundwork for future research on control in co-creative AI systems.

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